Source: `deliverables/PHASE12_DECISION_MEMO.md` (repository dewarjoseph/-)

Phase XII decision memo: what to run next, and why

Where things stand (certified)

- Over all 1 679 616 Chair44 frame rules, **the maximiser of E_* is one of 1387, 539311, 539291**, and the three intervals overlap:

rule	interval
539311	[9.2083494, 9.3817818]
1387	[9.1104981, 9.3616608]
539291	[7.1408386, 9.2483895]
1389 (excluded)	[8.7644993, 9.2083453], below the lower bound of 539311

  - The lower bound of 1387 (radius 20, 24 571 balls) is reproduced by an implementation sharing no code with the certifying one.
  - The lower bound of 539311 is the preregistered radius-24 certificate (39 117 balls, all certified), run by one implementation.
  - The upper bounds are exact rational Rayleigh quotients on the pinned level-7 supertiles (2 097 152 tiles).
- **The six-member S_3 orbit that topped the funnel is explained, and it is not a symmetry.**
  - The right action of the chair's point group is a gauge symmetry of one inflation step (Orbit Lemma).
  - It is a full symmetry on the conjugation stabiliser (and the closure of such moves), and it is certified broken in infinite volume for 1387 against 539288.

The previous decision, and what it returned

The last version of this memo chose **the level-7 frontier ordering test**, with a prediction frozen before any Y_7 value existed (`phase12_preregistration_next.json`):

1. the order 539311 > 1387 > 539291 > 1389 persists at L = 7;
2. $Y_7(539311) - Y_7(1387) \in [0.010, 0.030]$.

Both held: the order is unchanged, and the gap is 0.020121 (§14.1 of the report). As the memo said in advance, this is evidence [E], not proof, that **the extremiser is the mirror twin 539311, not 1387**.

A second protocol, frozen before the level-7 values and the new probes (`phase12_preregistration_extrapolati`) compared an extrapolation from above (Y_5–Y_7) with one from below (sampled radius-R probes). It reports **[X] evidence that E_*(539311) > E_*(1387) > E_*(539291), E_*(1389)**, and estimates the co-leaders' floors at 9.371–9.380 and 9.359–9.360 (§14.2).

The decision now

Run and passed: the radius-24 certificate of 539311 at $\mu = 9655658/2^{20} = 9.2083530$ (`phase12_preregistration_r24.json`, frozen before $Y_7(1389)$ and before any radius-24 data of 539311).

- Every one of the 39 117 legal radius-24 balls passed: $E_*(539311) \geq 9.2083494 > 9.2083453 \geq Y_7(1389) \geq E_*(1389)$. **1389 is proven inferior, and the frontier is three** (report §14.3).
- Go/no-go had passed: the 60-ball probe gave 9.2177320 > μ.
- Cost: 39 117 balls, about 13 h on 4 cores, resumed three times from its checkpoint after the container was reclaimed.

Next, if compute is available: the radius-28 certificate of 539311 (μ above $Y_7(539291)$ = 9.2483895; about 70 000 balls, 60 h on 4 cores). It would leave 1387 against 539311. It must be preregistered before it is run.

Why this, and not another

option	why not, or why
a certificate separating 1387 and 539311	needs radius 67–91 by the two models [X]: 10^6–10^7 balls and 10^6–10^7 core-hours. Out of reach, and Theorem C(iv) says no geometry-only method does better than R^{-2}
the radius-24 certificate of 539311	**chosen, and passed.** It is the cheapest certificate that changes the certified frontier, its outcome was not known when it was frozen, and it can fail (any ball below μ)
a radius-28 certificate of 539311, to exclude 539291 as well	the same idea one step further (μ above $Y_7(539291) = 9.2483895$); about 70 000 balls and 60 h on 4 cores. The natural follow-up if radius 24 passes
Y_8 (16.8 million tiles)	it would sharpen the extrapolation from above, but it is 8× the memory of Y_7 (over the 13.7 GB available) and could not change any certified statement
a new lower-bound method that beats R^{-2}	the only route to a theorem about the co-leaders, below. It has no concrete construction yet, so no falsifiable prediction can be frozen today

What each outcome means

- **The certificate passes (this happened).** E_*(539311) ≥ 9.2083494 [C], 1389 is excluded, and 539311 also becomes the rule with the highest certified lower bound. The question of the maximiser is then 1387 against 539311 (and 539291, until radius 28).
- **It fails.** Some legal radius-24 ball of 539311 lies below μ. The frontier stays at four. The failing balls are recorded, and they show where the probe and the lower model were too optimistic.

What none of this can do

No upper bound and no extrapolation certifies an ordering of floors. The question **is E_*(539311) > E_*(1387)?** stays open until a lower bound for 539311 exceeds 9.3616608.

The open bottleneck

A certificate of the form E_* ≥ $\mathcal{F} - \mathcal{E}$ with $\mathcal{E} = o(R^{-2})$ for one fixed operator family (Chair44). Theorem C(iv) shows this is impossible uniformly over connections. Whether it is possible for a single hierarchical hull is the specific unresolved proof obligation. The level-7 data give one hint where to look: the bottom of T_7 is a dense cluster (the lowest Ritz values lie within $2–7\cdot10^{-5}$ of one another), and for three of the four rules the successive differences Y_L − Y_{L+1} shrink by almost exactly a factor 4 per level (Aitken's q = 0.250–0.262). The low-energy states behave like those of a Dirichlet box with an effective kinetic constant κ_eff ≈ 3.3–3.5 [X], against ≈ 7.8 implied by the IMS fits. A localisation adapted to them would improve the constant by a factor of about 2.3, yet would still need R ≈ 46–62 for the co-leaders (§14.3). **What is needed is a better rate, not a better constant**: for instance a hierarchical (renormalisation) lower bound that compares T_{L+1} with its eight copies of T_L and controls the gluing error by something smaller than R^{-2}.